Guidance for Industry

Q10 Pharmaceutical Quality System

U.S. Department of Health and Human Services
Food and Drug Administration
Center for Drug Evaluation and Research (CDER)
Center for Biologics Evaluation and Research (CBER)

April 2009
ICH

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I. INTRODUCTION (1, 1.1)[2]

This internationally harmonized guidance is intended to assist pharmaceutical manufacturers by describing a model for an effective *quality* management system for the pharmaceutical industry, referred to as the *pharmaceutical quality system*. Throughout this guidance, the term *pharmaceutical quality system* refers to the ICH Q10 model.

ICH Q10 describes one comprehensive model for an effective pharmaceutical quality system that is based on International Organization for Standardization (ISO) quality concepts, includes applicable good manufacturing practice (GMP) regulations, and complements ICH "Q8 Pharmaceutical Development" and ICH "Q9 Quality Risk Management."[3] ICH Q10 is a model for a pharmaceutical quality system that can be implemented throughout the different stages of a product lifecycle. Much of the content of ICH Q10 applicable to manufacturing sites is currently specified by regional GMP requirements. ICH Q10 is not intended to create any new expectations beyond current regulatory requirements. Consequently, the content of ICH Q10 that is additional to current regional GMP requirements is optional.

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[1] This guidance was developed within the Expert Working Group (Quality) of the International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) and has been subject to consultation by the regulatory parties, in accordance with the ICH process. This document has been endorsed by the ICH Steering Committee at *Step 4* of the ICH process, June 2008. At *Step 4* of the process, the final draft is recommended for adoption to the regulatory bodies of the European Union, Japan, and the United States.

[2] Arabic numbers reflect the organizational breakdown of the document endorsed by the ICH Steering Committee at Step 4 of the ICH process, June 2008.

[3] These guidances are available on the Internet at http://www.fda.gov/cder/guidance/index.htm. We update guidances periodically. To make sure you have the most recent version of a guidance, check the CDER guidance page at http://www.fda.gov/cder/guidance/index.htm

ICH Q10 demonstrates industry and regulatory authorities' support of an effective pharmaceutical quality system to enhance the quality and availability of medicines around the world in the interest of public health. Implementation of ICH Q10 throughout the product lifecycle should facilitate *innovation* and *continual improvement* and strengthen the link between pharmaceutical development and manufacturing activities.

FDA's guidance documents, including this guidance, do not establish legally enforceable responsibilities. Instead, guidances describe the Agency's current thinking on a topic and should be viewed only as recommendations, unless specific regulatory or statutory requirements are cited. The use of the word *should* in Agency guidances means that something is suggested or recommended, but not required.

II. PHARMACEUTICAL QUALITY MANAGEMENT SYSTEM

A. Scope (1.2)

This guidance applies to the systems supporting the development and manufacture of pharmaceutical drug substances (i.e., active pharmaceutical ingredients (APIs)) and drug products, including biotechnology and biological products, throughout the product lifecycle.

The elements of ICH Q10 should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the differences among, and the different goals of, each stage (see section IV (3)).

For the purposes of this guidance, the product lifecycle includes the following technical activities for new and existing products:

- Pharmaceutical Development
  - Drug substance development
  - Formulation development (including container/closure system)
  - Manufacture of investigational products
  - Delivery system development (where relevant)
  - Manufacturing process development and scale-up
  - Analytical method development
- Technology Transfer
  - New product transfers during development through manufacturing
  - Transfers within or between manufacturing and testing sites for marketed products
- Commercial Manufacturing
  - Acquisition and control of materials
  - Provision of facilities, utilities, and equipment
  - Production (including packaging and labeling)
  - Quality control and assurance
  - Release
  - Storage

- Distribution (excluding wholesaler activities)

- Product Discontinuation
  - Retention of documentation
  - Sample retention
  - Continued product assessment and reporting

B. Relationship of ICH Q10 to Regional GMP Requirements, ISO Standards, and ICH Q7 (1.3)

Regional GMP requirements, the ICH guidance "Q7 Good Manufacturing Practice Guidance for Active Pharmaceutical Ingredients," and ISO quality management system guidelines form the foundation for ICH Q10. To meet the objectives described below, ICH Q10 augments GMPs by describing specific quality system elements and management responsibilities. ICH Q10 provides a harmonized model for a pharmaceutical quality system throughout the lifecycle of a product and is intended to be used together with regional GMP requirements.

The regional GMPs do not explicitly address all stages of the product lifecycle (e.g., development). The quality system elements and management responsibilities described in this guidance are intended to encourage the use of science- and risk-based approaches at each lifecycle stage, thereby promoting continual improvement across the entire product lifecycle.

C. Relationship of ICH Q10 to Regulatory Approaches (1.4)

Regulatory approaches for a specific product or manufacturing facility should be commensurate with the level of product and process understanding, the results of *quality risk management*, and the effectiveness of the pharmaceutical quality system. When implemented, the effectiveness of the pharmaceutical quality system can normally be evaluated during a regulatory inspection at the manufacturing site. Potential opportunities to enhance science- and risk-based regulatory approaches are identified in Annex 1. Regulatory processes will be determined by region.

D. ICH Q10 Objectives (1.5)

Implementation of the Q10 model should result in achievement of three main objectives that complement or enhance regional GMP requirements.

1. Achieve ***Product Realization*** *(1.5.1)*

To establish, implement, and maintain a system that allows the delivery of products with the quality attributes appropriate to meet the needs of patients, health care professionals, regulatory authorities (including compliance with approved regulatory filings) and other internal and external customers.

2 Establish and Maintain a ***State of Control*** *(1.5.2)*

To develop and use effective monitoring and control systems for process performance and product quality, thereby providing assurance of continued suitability and *capability of processes*. Quality risk management can be useful in identifying the monitoring and control systems.

3 Facilitate Continual Improvement (1.5.3)

To identify and implement appropriate product quality improvements, process improvements, variability reduction, innovations, and pharmaceutical quality system enhancements, thereby increasing the ability to fulfill a pharmaceutical manufacturer's own quality needs consistently. Quality risk management can be useful for identifying and prioritizing areas for continual improvement.

E. *Enablers*: Knowledge Management and Quality Risk Management (1.6)

Use of *knowledge management* and quality risk management will enable a company to implement ICH Q10 effectively and successfully. These enablers will facilitate achievement of the objectives described in section II.D (1.5) above by providing the means for science- and risk-based decisions related to product quality.

1. Knowledge Management (1.6.1)

Product and process knowledge should be managed from development through the commercial life of the product up to and including product discontinuation. For example, development activities using scientific approaches provide knowledge for product and process understanding. Knowledge management is a systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components. Sources of knowledge include, but are not limited to, prior knowledge (public domain or internally documented); pharmaceutical development studies; technology transfer activities; process validation studies over the product lifecycle; manufacturing experience; innovation; continual improvement; and *change management* activities.

2. Quality Risk Management (1.6.2)

Quality risk management is integral to an effective pharmaceutical quality system. It can provide a proactive approach to identifying, scientifically evaluating, and controlling potential risks to quality. It facilitates continual improvement of process performance and product quality throughout the product lifecycle. ICH Q9 provides principles and examples of tools for quality risk management that can be applied to different aspects of pharmaceutical quality.

F. Design and Content Considerations (1.7)

(a) The design, organization, and documentation of the pharmaceutical quality system should be well structured and clear to facilitate common understanding and consistent application.

(b) The elements of ICH Q10 should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the different goals and knowledge available for each stage.

(c) The size and complexity of the company's activities should be taken into consideration when developing a new pharmaceutical quality system or modifying an existing one. The design of the pharmaceutical quality system should incorporate appropriate risk management principles. While some aspects of the pharmaceutical quality system can be company wide and others site specific, the effectiveness of the pharmaceutical quality system is normally demonstrated at the site level.

(d) The pharmaceutical quality system should include appropriate processes, resources, and responsibilities to provide assurance of the quality of *outsourced activities* and purchased materials as described in section III.G (2.7).

(e) Management responsibilities, as described in section III (2), should be identified within the pharmaceutical quality system.

(f) The pharmaceutical quality system should include the following elements, as described in section IV (3): process performance and product quality monitoring, *corrective* and *preventive action*, change management, and management review.

(g) *Performance indicators*, as described in section V (4), should be identified and used to monitor the effectiveness of processes within the pharmaceutical quality system.

G. Quality Manual (1.8)

A *Quality Manual* or equivalent documentation approach should be established and should contain the description of the pharmaceutical quality system. The description should include:

(a) The *quality policy* (see section III (2)).

(b) The scope of the pharmaceutical quality system.

(c) Identification of the pharmaceutical quality system processes, as well as their sequences, linkages, and interdependencies. Process maps and flow charts can be useful tools to facilitate depicting pharmaceutical quality system processes in a visual manner.

(d) Management responsibilities within the pharmaceutical quality system (see section III (2)).

III. MANAGEMENT RESPONSIBILITY (2)

Leadership is essential to establish and maintain a company-wide commitment to quality and for the performance of the pharmaceutical quality system.

A. Management Commitment (2.1)

(a) *Senior management* has the ultimate responsibility to ensure an effective pharmaceutical quality system is in place to achieve the *quality objectives,* and that roles, responsibilities, and authorities are defined, communicated, and implemented throughout the company.

(b) Management should:

(1) Participate in the design, implementation, monitoring, and maintenance of an effective pharmaceutical quality system.

(2) Demonstrate strong and visible support for the pharmaceutical quality system and ensure its implementation throughout their organization.

(3) Ensure a timely and effective communication and escalation process exists to raise quality issues to the appropriate levels of management.

(4) Define individual and collective roles, responsibilities, authorities, and inter-relationships of all organizational units related to the pharmaceutical quality system. Ensure these interactions are communicated and understood at all levels of the organization. An independent quality unit/structure with authority to fulfill certain pharmaceutical quality system responsibilities is required by regional regulations.

(5) Conduct management reviews of process performance and product quality and of the pharmaceutical quality system.

(6) Advocate continual improvement.

(7) Commit appropriate resources.

B. Quality Policy (2.2)

(a) Senior management should establish a quality policy that describes the overall intentions and direction of the company related to quality.

(b) The quality policy should include an expectation to comply with applicable regulatory requirements and should facilitate continual improvement of the pharmaceutical quality system.

(c) The quality policy should be communicated to and understood by personnel at all levels in the company.

(d) The quality policy should be reviewed periodically for continuing effectiveness.

C. *Quality Planning (*2.3)

(a) Senior management should ensure the quality objectives to implement the quality policy are defined and communicated.

(b) Quality objectives should be supported by all relevant levels of the company.

(c) Quality objectives should align with the company's strategies and be consistent with the quality policy.

(d) Management should provide the appropriate resources and training to achieve the quality objectives.

(e) Performance indicators that measure progress against quality objectives should be established, monitored, communicated regularly, and acted upon as appropriate as described in section V.A (4.1) of this document.

D. Resource Management (2.4)

(a) Management should determine and provide adequate and appropriate resources (human, financial, materials, facilities, and equipment) to implement and maintain the pharmaceutical quality system and continually improve its effectiveness.

(b) Management should ensure that resources are appropriately applied to a specific product, process, or site.

E. Internal Communication (2.5)

(a) Management should ensure appropriate communication processes are established and implemented within the organization.

(b) Communications processes should ensure the flow of appropriate information between all levels of the company.

(c) Communication processes should ensure the appropriate and timely escalation of certain product quality and pharmaceutical quality system issues.

F. Management Review (2.6)

(a) Senior management should be responsible for pharmaceutical quality system governance through management review to ensure its continuing suitability and effectiveness.

(b) Management should assess the conclusions of periodic reviews of process performance and product quality and of the pharmaceutical quality system, as described in sections IV (3) and V (4).

G. Management of Outsourced Activities and Purchased Materials (2.7)

The pharmaceutical quality system, including the management responsibilities described in this section, extends to the control and review of any outsourced activities and quality of purchased materials. The pharmaceutical company is ultimately responsible to ensure processes are in place to assure the control of outsourced activities and quality of purchased materials. These processes should incorporate quality risk management and include:

(a) Assessing prior to outsourcing operations or selecting material suppliers, the suitability and competence of the other party to carry out the activity or provide the material using a defined supply chain (e.g., audits, material evaluations, qualification).

(b) Defining the responsibilities and communication processes for quality-related activities of the involved parties. For outsourced activities, this should be included in a written agreement between the contract giver and contract acceptor.

(c) Monitoring and review of the performance of the contract acceptor or the quality of the material from the provider, and the identification and implementation of any essential improvements.

(d) Monitoring incoming ingredients and materials to ensure they are from approved sources using the agreed supply chain.

H. Management of Change in Product Ownership (2.8)

When product ownership changes (e.g., through acquisitions), management should consider the complexity of this and ensure:

(a) The ongoing responsibilities are defined for each company involved

(b) The essential information is transferred

IV. CONTINUAL IMPROVEMENT OF PROCESS PERFORMANCE AND PRODUCT QUALITY (3)

This section describes the lifecycle stage goals and the four specific pharmaceutical quality system elements that augment regional requirements to achieve the ICH Q10 objectives, as defined in section II.D (1.5). It does not restate all regional GMP requirements.

A. Lifecycle Stage Goals (3.1)

The goals of each product lifecycle stage are described below.

1. Pharmaceutical Development (3.1.1)

The goal of pharmaceutical development activities is to design a product and its manufacturing process to consistently deliver the intended performance and meet the needs of patients and healthcare professionals, and regulatory authorities and internal customers' requirements. Approaches to pharmaceutical development are described in ICH Q8. The results of exploratory and clinical development studies, while outside the scope of this guidance, are inputs to pharmaceutical development.

2. Technology Transfer (3.1.2)

The goal of technology transfer activities is to transfer product and process knowledge between development and manufacturing, and within or between manufacturing sites to achieve product realization. This knowledge forms the basis for the manufacturing process, *control strategy*, process validation approach, and ongoing continual improvement.

3. Commercial Manufacturing (3.1.3)

The goals of manufacturing activities include achieving product realization, establishing and maintaining a state of control, and facilitating continual improvement. The pharmaceutical quality system should assure that the desired product quality is routinely met, suitable process performance is achieved, the set of controls are appropriate, improvement opportunities are identified and evaluated, and the body of knowledge is continually expanded.

4. Product Discontinuation (3.1.4)

The goal of product discontinuation activities is to manage the terminal stage of the product lifecycle effectively. For product discontinuation, a predefined approach should be used to manage activities such as retention of documentation and samples and continued product assessment (e.g., complaint handling and stability) and reporting in accordance with regulatory requirements.

B. Pharmaceutical Quality System Elements (3.2)

The elements described below might be required in part under regional GMP regulations. However, the Q10 model's intent is to enhance these elements to promote the lifecycle approach to product quality. These four elements are:

- Process performance and product quality monitoring system
- *Corrective action* and *preventive action* (CAPA) system
- Change management system
- Management review of process performance and product quality

These elements should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the differences among the stages and the different goals of each stage. Throughout the product lifecycle, companies are encouraged to evaluate opportunities for innovative approaches to improve product quality.

Each element is followed by a table of example applications of the element to the stages of the pharmaceutical lifecycle.

1. Process Performance and Product Quality Monitoring System 3.2.1

Pharmaceutical companies should plan and execute a system for the monitoring of process performance and product quality to ensure a state of control is maintained. An effective monitoring system provides assurance of the continued capability of processes and controls to produce a product of desired quality and to identify areas for continual improvement. The process performance and product quality monitoring system should:

(a) Use quality risk management to establish the control strategy. This can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control. The control strategy should facilitate timely *feedback/feedforward* and appropriate corrective action and preventive action.

(b) Provide the tools for measurement and analysis of parameters and attributes identified in the control strategy (e.g., data management and statistical tools).

(c) Analyze parameters and attributes identified in the control strategy to verify continued operation within a state of control.

(d) Identify sources of variation affecting process performance and product quality for potential continual improvement activities to reduce or control variation.

(e) Include feedback on product quality from both internal and external sources (e.g., complaints, product rejections, nonconformances, recalls, deviations, audits and regulatory inspections, and findings).

(f) Provide knowledge to enhance process understanding, enrich the *design space* (where established), and enable innovative approaches to process validation.

Table I: Application of Process Performance and Product Quality Monitoring System Throughout the Product Lifecycle

Pharmaceutical Development	**Technology Transfer**	**Commercial Manufacturing**	**Product Discontinuation**
Process and product knowledge generated and process and product monitoring conducted throughout development can be used to establish a control strategy for manufacturing.	Monitoring during scale-up activities can provide a preliminary indication of process performance and the successful integration into manufacturing. Knowledge obtained during transfer and scale-up activities can be useful in further developing the control strategy.	A well-defined system for process performance and product quality monitoring should be applied to assure performance within a state of control and to identify improvement areas.	Once manufacturing ceases, monitoring such as stability testing should continue to completion of the studies. Appropriate action on marketed product should continue to be executed according to regional regulations.

2. Corrective Action and Preventive Action (CAPA) System (3.2.2)

The pharmaceutical company should have a system for implementing corrective actions and preventive actions resulting from the investigation of complaints, product rejections, nonconformances, recalls, deviations, audits, regulatory inspections and findings, and trends from process performance and product quality monitoring. A structured approach to the investigation process should be used with the objective of determining the root cause. The level of effort, formality, and documentation of the investigation should be commensurate with the level of risk, in line with ICH Q9. CAPA methodology should result in product and process improvements and enhanced product and process understanding.

Table II: Application of Corrective Action and Preventive Action System Throughout the Product Lifecycle

Pharmaceutical Development	**Technology Transfer**	**Commercial Manufacturing**	**Product Discontinuation**
Product or process variability is explored. CAPA methodology is useful where corrective actions and preventive actions are incorporated into the iterative design and development process.	CAPA can be used as an effective system for feedback, feedforward, and continual improvement.	CAPA should be used, and the effectiveness of the actions should be evaluated.	CAPA should continue after the product is discontinued. The impact on product remaining on the market should be considered, as well as other products that might be affected.

3. Change Management System (3.2.3)

Innovation, continual improvement, the outputs of process performance and product quality monitoring, and CAPA drive change. To evaluate, approve, and implement these changes properly, a company should have an effective change management system. There is generally a difference in formality of change management processes prior to the initial regulatory submission and after submission, where changes to the regulatory filing might be required under regional requirements.

The change management system ensures continual improvement is undertaken in a timely and effective manner. It should provide a high degree of assurance there are no unintended consequences of the change.

The change management system should include the following, as appropriate for the stage of the lifecycle:

(a) Quality risk management should be utilized to evaluate proposed changes. The level of effort and formality of the evaluation should be commensurate with the level of risk.

(b) Proposed changes should be evaluated relative to the marketing authorization, including design space, where established, and/or current product and process understanding. There should be an assessment to determine whether a change to the regulatory filing is required under regional requirements. As stated in ICH Q8, working within the design space is not considered a change (from a regulatory filing perspective). However, from a pharmaceutical quality system standpoint, all changes should be evaluated by a company’s change management system.

(c) Proposed changes should be evaluated by expert teams contributing the appropriate expertise and knowledge from relevant areas (e.g., Pharmaceutical Development, Manufacturing, Quality, Regulatory Affairs, and Medical) to ensure the change is technically justified. Prospective evaluation criteria for a proposed change should be set.

(d) After implementation, an evaluation of the change should be undertaken to confirm the change objectives were achieved and that there was no deleterious impact on product quality.

Table III: Application of Change Management System Throughout the Product Lifecycle

Pharmaceutical Development	**Technology Transfer**	**Commercial Manufacturing**	**Product Discontinuation**
Change is an inherent part of the development process and should be documented; the formality of the change management process should be consistent with the stage of pharmaceutical development.	The change management system should provide management and documentation of adjustments made to the process during technology transfer activities.	A formal change management system should be in place for commercial manufacturing. Oversight by the quality unit should provide assurance of appropriate science- and risk-based assessments.	Any changes after product discontinuation should go through an appropriate change management system.

4. Management Review of Process Performance and Product Quality (3.2.4)

Management review should provide assurance that process performance and product quality are managed over the lifecycle. Depending on the size and complexity of the company, management review can be a series of reviews at various levels of management and should include a timely and effective communication and escalation process to raise appropriate quality issues to senior levels of management for review.

(a) The management review system should include:

- (1) The results of regulatory inspections and findings, audits and other assessments, and commitments made to regulatory authorities
- (2) Periodic quality reviews, that can include:
  - (i) Measures of customer satisfaction such as product quality complaints and recalls
  - (ii) Conclusions of process performance and product quality monitoring
  - (iii) The effectiveness of process and product changes including those arising from corrective action and preventive actions
- (3) Any follow-up actions from previous management reviews

(b) The management review system should identify appropriate actions, such as:

- (1) Improvements to manufacturing processes and products
- (2) Provision, training, and/or realignment of resources

(3) Capture and dissemination of knowledge

Table IV: Application of Management Review of Process Performance and Product Quality Throughout the Product Lifecycle

Pharmaceutical Development	**Technology Transfer**	**Commercial Manufacturing**	**Product Discontinuation**
Aspects of management review can be performed to ensure adequacy of the product and process design.	Aspects of management review should be performed to ensure the developed product and process can be manufactured at commercial scale.	Management review should be a structured system, as described above, and should support continual improvement.	Management review should include such items as product stability and product quality complaints.

V. CONTINUAL IMPROVEMENT OF THE PHARMACEUTICAL QUALITY SYSTEM (4)

This section describes activities that should be conducted to manage and continually improve the pharmaceutical quality system.

A. Management Review of the Pharmaceutical Quality System (4.1)

Management should have a formal process for reviewing the pharmaceutical quality system on a periodic basis. The review should include:

(a) Measurement of achievement of pharmaceutical quality system objectives

(b) Assessment of performance indicators that can be used to monitor the effectiveness of processes within the pharmaceutical quality system, such as:

(1) Complaint, deviation, CAPA and change management processes
(2) Feedback on outsourced activities
(3) Self-assessment processes including risk assessments, trending, and audits
(4) External assessments such as regulatory inspections and findings and customer audits

B. Monitoring of Internal and External Factors That Can Have an Impacton the Pharmaceutical Quality System (4.2)

Factors monitored by management can include:

(a) Emerging regulations, guidance, and quality issues that can have an impact on the Pharmaceutical Quality System

(b) Innovations that might enhance the pharmaceutical quality system

(c) Changes in business environment and objectives

(d) Changes in product ownership

C. Outcomes of Management Review and Monitoring (4.3)

The outcome of management review of the pharmaceutical quality system and monitoring of internal and external factors can include:

(a) Improvements to the pharmaceutical quality system and related processes

(b) Allocation or reallocation of resources and/or personnel training

(c) Revisions to quality policy and quality objectives

(d) Documentation and timely and effective communication of the results of the management review and actions, including escalation of appropriate issues to senior management

VI. GLOSSARY (5)

ICH and ISO definitions are used in ICH Q10 where they exist. For the purpose of ICH Q10, where the words “requirement”, “requirements,” or “necessary” appear in an ISO definition, they do not necessarily reflect a regulatory requirement. The source of the definition is identified in parentheses after the definition. Where no appropriate ICH or ISO definition was available, an ICH Q10 definition was developed.

Capability of a Process: Ability of a process to realize a product that will fulfill the requirements of that product. The concept of process capability can also be defined in statistical terms. (ISO 9000:2005)

Change Management: A systematic approach to proposing, evaluating, approving, implementing, and reviewing changes. (ICH Q10)

Continual Improvement: Recurring activity to increase the ability to fulfill requirements. (ISO 9000:2005)

Control Strategy: A planned set of controls, derived from current product and process understanding, that assures process performance and product quality. The controls can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control. (ICH Q10)

Corrective Action: Action to eliminate the cause of a detected nonconformity or other undesirable situation. NOTE: Corrective action is taken to prevent recurrence whereas preventive action is taken to prevent occurrence. (ISO 9000:2005)

Design Space: The multidimensional combination and interaction of input variables (e.g., material attributes) and process parameters that have been demonstrated to provide assurance of quality. (ICH Q8)

Enabler: A tool or process that provides the means to achieve an objective. (ICH Q10)

Feedback/Feedforward:

> **Feedback:** The modification or control of a process or system by its results or effects.
>
> **Feedforward:** The modification or control of a process using its anticipated results or effects. (Oxford Dictionary of English by Oxford University Press, 2003)

Feedback/feedforward can be applied technically in process control strategies and conceptually in quality management. (ICH Q10)

Innovation: The introduction of new technologies or methodologies. (ICH Q10)

Knowledge Management: Systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components. (ICH Q10)

Outsourced Activities: Activities conducted by a contract acceptor under a written agreement with a contract giver. (ICH Q10)

Performance Indicators: Measurable values used to quantify quality objectives to reflect the performance of an organization, process, or system, also known as *performance metrics* in some regions. (ICH Q10)

Pharmaceutical Quality System (PQS): Management system to direct and control a pharmaceutical company with regard to quality. (ICH Q10 based upon ISO 9000:2005)

Preventive Action: Action to eliminate the cause of a potential nonconformity or other undesirable potential situation. NOTE: Preventive action is taken to prevent occurrence whereas corrective action is taken to prevent recurrence. (ISO 9000:2005)

Product Realization: Achievement of a product with the quality attributes appropriate to meet the needs of patients, health care professionals, and regulatory authorities (including compliance with marketing authorization) and internal customers requirements. (ICH Q10)

Quality: The degree to which a set of inherent properties of a product, system, or process fulfils requirements. (ICH Q9)

Quality Manual: Document specifying the quality management system of an organization. (ISO 9000:2005)

Quality Objectives: A means to translate the quality policy and strategies into measurable activities. (ICH Q10)

Quality Planning: Part of quality management focused on setting quality objectives and specifying necessary operational processes and related resources to fulfill the quality objectives. (ISO 9000:2005)

Quality Policy: Overall intentions and direction of an organization related to quality as formally expressed by senior management. (ISO 9000:2005)

Quality Risk Management: A systematic process for the assessment, control, communication, and review of risks to the quality of the drug (medicinal) product across the product lifecycle. (ICH Q9)

Senior Management: Person(s) who direct and control a company or site at the highest levels with the authority and responsibility to mobilize resources within the company or site. (ICH Q10 based in part on ISO 9000:2005)

State of Control: A condition in which the set of controls consistently provides assurance of continued process performance and product quality. (ICH Q10)

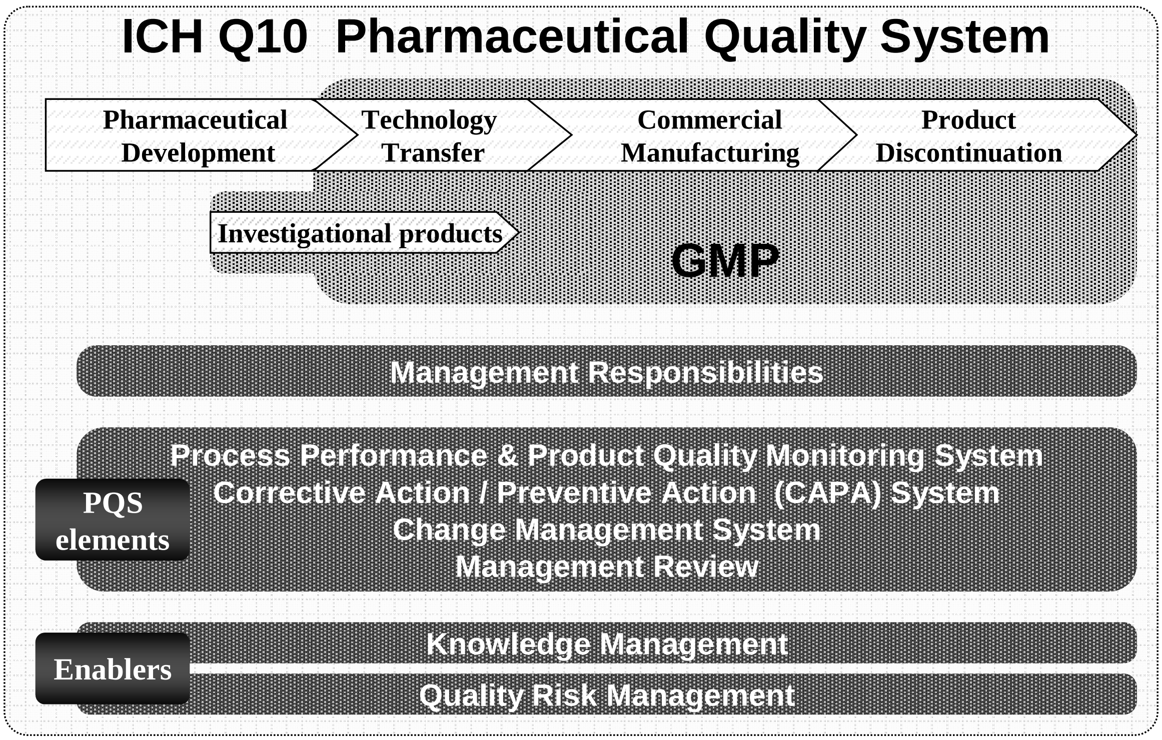
Annex I: Potential Opportunities To Enhance Science- and Risk-Based Regulatory Approaches *

*Note: This annex reflects potential opportunities to enhance regulatory approaches. The actual regulatory process will be determined by region.

Scenario	Potential Opportunity
1. Comply with GMPs	Compliance – status quo
2. Demonstrate effective pharmaceutical quality system, including effective use of quality risk management principles (e.g., ICH Q9 and ICH Q10).	Opportunity to: • increase use of risk-based approaches for regulatory inspections
3. Demonstrate product and process understanding, including effective use of quality risk management principles (e.g., ICH Q8 and ICH Q9).	Opportunity to: • facilitate science-based pharmaceutical quality assessment • enable innovative approaches to process validation • establish real-time release mechanisms
4. Demonstrate effective pharmaceutical quality system and product and process understanding, including the use of quality risk management principles (e.g., ICH Q8, ICH Q9, and ICH Q10).	Opportunity to: • increase use of risk-based approaches for regulatory inspections; • facilitate science-based pharmaceutical quality assessment; • optimize science- and risk-based postapproval change processes to maximize benefits from innovation and continual improvement; • enable innovative approaches to process validation; • establish real-time release mechanisms.

Annex 2

Diagram of the ICH Q10 Pharmaceutical Quality System Model



This diagram illustrates the major features of the ICH Q10 Pharmaceutical Quality System (PQS) model. The PQS covers the entire lifecycle of a product including pharmaceutical development, technology transfer, commercial manufacturing, and product discontinuation as illustrated by the upper portion of the diagram. The PQS augments regional GMPs as illustrated in the diagram. The diagram also illustrates that regional GMPs apply to the manufacture of investigational products.

The next horizontal bar illustrates the importance of management responsibilities explained in section III (2) to all stages of the product lifecycle. The following horizontal bar lists the PQS elements that serve as the major pillars under the PQS model. These elements should be applied appropriately and proportionally to each lifecycle stage, recognizing opportunities to identify areas for continual improvement.

The bottom set of horizontal bars illustrates the enablers: knowledge management and quality risk management, which are applicable throughout the lifecycle stages. These enablers support the PQS goals of achieving product realization, establishing and maintaining a state of control, and facilitating continual improvement.

Annex 4

Guideline on data integrity

This document replaces the WHO *Guidance on good data and record management practices* (Annex 5, WHO Technical Report Series, No. 996, 2016) (*1*).

1. Introduction and background

1.1. In recent years, the number of observations made regarding the integrity of data, documentation and record management practices during inspections of good manufacturing practice (GMP) (*2*), good clinical practice (GCP), good laboratory practice (GLP) and Good Trade and Distribution Practices (GTDP) have been increasing. The possible causes for this may include (i) reliance on inadequate human practices; (ii) poorly defined procedures; (iii) resource constraints; (iv) the use of computerized systems that are not capable of meeting regulatory requirements or are inappropriately managed and validated (*3*, *4*); (v) inappropriate and inadequate control of data flow; and (vi) failure to adequately review and manage original data and records.

1.2. Data governance and related measures should be part of a quality system, and are important to ensure the reliability of data and records in good practice (GxP) activities and regulatory submissions. The data and records should be 'attributable, legible, contemporaneous, original' and accurate, complete, consistent, enduring, and available; commonly referred to as "ALCOA+".

1.3. This document replaces the WHO *Guidance on good data and record management practices* (Annex 5, WHO Technical Report Series, No. 996, 2016) (*1*).

2. Scope

2.1. This document provides information, guidance and recommendations to strengthen data integrity in support of product quality, safety and efficacy. The aim is to ensure compliance with regulatory requirements in, for example clinical research, production and quality control, which ultimately contributes to patient safety. It covers electronic, paper and hybrid systems.

2.2. The guideline covers "GxP" for medical products. The principles could also be applied to other products such as vector control products.

2.3. The principles of this guideline also apply to contract givers and contract acceptors. Contract givers are ultimately responsible for the integrity of data provided to them by contract acceptors. Contract givers should therefore ensure that contract acceptors have the appropriate capabilities and comply with the principles contained in this guideline and documented in quality agreements.

2.4. Where possible, this guideline has been harmonised with other published documents on data integrity. This guideline should also be read with other WHO good practices guidelines and publications including, but not limited to, those listed in the references section of this document.

3. Glossary

The definitions given below apply to the terms used in these guidelines. They may have different meanings in other contexts.

ALCOA+. A commonly used acronym for “attributable, legible, contemporaneous, original and accurate” which puts additional emphasis on the attributes of being complete, consistent, enduring and available throughout the data life cycle for the defined retention period.

Archiving. Archiving is the process of long-term storage and protection of records from the possibility of deterioration, and being altered or deleted, throughout the required retention period. Archived records should include the complete data, for example, paper records, electronic records including associated metadata such as audit trails and electronic signatures. Within a GLP context, the archived records should be under the control of independent data management personnel throughout the required retention period.

Audit trail. The audit trail is a form of metadata containing information associated with actions that relate to the creation, modification or deletion of GxP records. An audit trail provides for a secure recording of life cycle details such as creation, additions, deletions or alterations of information in a record, either paper or electronic, without obscuring or overwriting the original record. An audit trail facilitates the reconstruction of the history of such events relating to the record regardless of its medium, including the “who, what, when and why” of the action.

Backup. The copying of live electronic data, at defined intervals, in a secure manner to ensure that the data are available for restoration.

Certified true copy or true copy. A copy (irrespective of the type of media used) of the original record that has been verified (i.e. by a dated signature or by generation through a validated process) to have the same information, including data that describe the context, content, and structure, as the original.

Data. All original records and true copies of original records, including source data and metadata, and all subsequent transformations and reports of these data which are generated or recorded at the time of the GMP activity and which allow full and complete reconstruction and evaluation of the GMP activity. Data should be accurately recorded by permanent means at the time of the activity. Data may be contained in paper records (such as worksheets and logbooks), electronic records and audit trails, photographs, microfilm or microfiche, audio or video files or any other media whereby information related to GMP activities is recorded.

Data criticality. This is defined by the importance of the data for the quality and safety of the product and how important data are for a quality decision within production or quality control.

Data governance. The sum total of arrangements which provide assurance of data quality. These arrangements ensure that data, irrespective of the process, format or technology in which it is generated, recorded, processed, retained, retrieved and used will ensure an attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available record throughout the data life cycle.

Data integrity risk assessment (DIRA). The process to map out procedures, systems and other components that generate or obtain data; to identify and assess risks and implement appropriate controls to prevent or minimize lapses in the integrity of the data.

Data life cycle. All phases of the process by which data are created, recorded, processed, reviewed, analysed and reported, transferred, stored and retrieved and monitored, until retirement and disposal. There should be a planned approach to assessing, monitoring and managing the data and the risks to those data, in a manner commensurate with the potential impact on patient safety, product quality and/or the reliability of the decisions made throughout all phases of the data life cycle.

Dynamic data. Dynamic formats, such as electronic records, allow an interactive relationship between the user and the record content. For example, electronic records in database formats allow the user to track, trend and query data; chromatography records maintained as electronic records allow the user or reviewer (with appropriate access permissions) to reprocess the data and expand the baseline to view the integration more clearly.

Electronic signatures. A signature in digital form (bio-metric or non-biometric) that represents the signatory. In legal terms, it is the equivalent of the handwritten signature of the signatory.

Good practices (GxP). An acronym for the group of good practice guides governing the preclinical, clinical, manufacturing, testing, storage, distribution and post-market activities for regulated pharmaceuticals, biologicals and medical devices, such as GLP, GCP, GMP, good pharmacovigilance practices (GVP) and good distribution practices (GDP).

Hybrid system. The use of a combination of electronic systems and paper systems.

Medical product. A term that includes medicines, vaccines, diagnostics and medical devices.

Metadata. Metadata are data that provide the contextual information required to understand other data. These include structural and descriptive metadata, which describe the structure, data elements, interrelationships and other characteristics of data. They also permit data to be attributable to an individual. Metadata that are necessary to evaluate the meaning of data should be securely linked to the data and subject to adequate review. For example, in the measurement of weight, the number 8 is meaningless without metadata, such as, the unit, milligram, gram, kilogram, and so on. Other examples of metadata include the time or date stamp of an activity, the operator identification (ID) of the person who performed an activity, the instrument ID used, processing parameters, sequence files, audit trails and other data required to understand data and reconstruct activities.

Raw data. The original record (data) which can be described as the first-capture of information, whether recorded on paper or electronically. Raw data is synonymous with source data.

Static data. A static record format, such as a paper or electronic record, that is fixed and allows little or no interaction between the user and the record content. For example, once printed or converted to static electronic format chromatography records lose the capability of being reprocessed or enabling more detailed viewing of baseline.

4. Data governance

4.1. There should be a written policy on data integrity.

4.2. Senior management should be accountable for the implementation of systems and procedures in order to minimise the potential risk to data integrity, and to identify the residual risk using risk management techniques such as the principles of the guidance on quality risk management from WHO (*5*) and The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) (*6*).

4.3. Senior management is responsible for the establishment, implementation and control of an effective data governance system. Data governance should be embedded in the quality system. The necessary policies, procedures, training, monitoring and other systems should be implemented.

4.4. Data governance should ensure the application of ALCOA+ principles.

4.5. Senior management is responsible for providing the environment to establish, maintain and continually improve the quality culture, supporting the transparent and open reporting of deviations, errors or omissions and data integrity lapses at all levels of the organization. Appropriate, immediate action should be taken when falsification of data is identified. Significant lapses in data integrity that may impact patient safety, product quality or efficacy should be reported to the relevant medicine regulatory authorities.

4.6. The quality system, including documentation such as procedures and formats for recording and reviewing of data, should be appropriately designed and implemented in order to provide assurance that records and data meet the principles contained in this guideline.

4.7. Data governance should address the roles, responsibilities, accountability and define the segregation of duties throughout the life cycle and consider the design, operation and monitoring of processes/systems to comply with the principles of data integrity, including control over authorized and unauthorized changes to data.

4.8. Data governance control strategies using quality risk management (QRM) principles (5) are required to prevent or mitigate risks. The control strategy should aim to implement appropriate technical, organizational and procedural controls. Examples of controls may include, but are not limited to:

- the establishment and implementation of procedures that will facilitate compliance with data integrity requirements and expectations;
- the adoption of a quality culture within the company that encourages personnel to be transparent about failures, which includes a reporting mechanism inclusive of investigation and follow-up processes;
- the implementation of appropriate controls to eliminate or reduce risks to an acceptable level throughout the life cycle of the data;
- ensuring sufficient time and resources are available to implement and complete a data integrity programme; to monitor compliance

with data integrity policies, procedures and processes through e.g. audits and self-inspections; and to facilitate continuous improvement of both;
- the assignment of qualified and trained personnel and provision of regular training for personnel in, for example, GxP, and the principles of data integrity in computerized systems and manual/paper based systems;
- the implementation and validation of computerized systems appropriate for their intended use, including all relevant data integrity requirements in order to ensure that the computerized system has the necessary controls to protect the electronic data (*3*); and
- the definition and management of the appropriate roles and responsibilities for contract givers and contract acceptors, entered into quality agreements and contracts including a focus on data integrity requirements.

4.9. Data governance systems should include, for example:

- the creation of an appropriate working environment;
- active support of continual improvement in particular based on collecting feedback; and
- review of results, including the reporting of errors, unauthorized changes, omissions and undesirable results.

4.10. The data governance programme should include policies and procedures addressing data management. These should at least where applicable, include:

- management oversight and commitment;
- the application of QRM;
- compliance with data protection legislation and best practices;
- qualification and validation policies and procedures;
- change, incident and deviation management;
- data classification, confidentiality and privacy;
- security, cybersecurity, access and configuration control;
- database build, data collection, data review, blinded data, randomization;
- the tracking, trending, reporting of data integrity anomalies, and lapses or failures for further action;

- the prevention of commercial, political, financial and other organizational pressures;
- adequate resources and systems;
- workload and facilities to facilitate the right environment that supports DI and effective controls;
- monitoring;
- record-keeping;
- training; and
- awareness of the importance of data integrity, product quality and patient safety.

4.11. There should be a system for the regular review of data for consistency with ALCOA+ principles. This includes paper records and electronic records in day-to-day work, system and facility audits and self-inspections.

4.12. The effort and resources applied to assure the integrity of the data should be commensurate with the risk and impact of a data integrity failure.

4.13. Where weaknesses in data integrity are identified, the appropriate corrective and preventive actions (CAPA) should be implemented across all relevant activities and systems and not in isolation.

4.14. Changing from paper-based systems to automated or computerised systems (or vice-versa) will not in itself remove the need for appropriate data integrity controls.

4.15. Records (paper and electronic) should be kept in a manner that ensures compliance with the principles of this guideline. These include but are not limited to:

- ensuring time accuracy of the system generating the record, accurately configuring and verifying time zone and time synchronisation, and restricting the ability to change dates, time zones and times for recording events;
- using controlled documents and forms for recording GxP data;
- defining access and privilege rights to GxP automated and computerized systems, ensuring segregation of duties;
- ensuring audit trail activation for all interactions and restricting the ability to enable or disable audit trails (Note: 'back-end' changes and 'hard' changes, such as hard deletes, should not be allowed). Where audit trials can be disabled then this this action should also appear in the audit trail;

- having automated data capture systems and printers connected to equipment and instruments in production (such as Supervisory Control and Data Acquisition (SCADA), Human Machine Interface (HMI) and Programme Logic Control (PLCs) systems), in , quality control, and in clinical research (such as Clinical Data Management (CDM) systems), where possible;
- designing processes in a way to avoid the unnecessary transcription of data or unnecessary conversion from paper to electronic and vice versa; and
- ensuring the proximity of an official GxP time source to site of GxP activity and record creation.

4.16. Systems, procedures and methodology used to record and store data should be periodically reviewed for effectiveness. These should be updated throughout the data life cycle, as necessary, where new technology becomes available. New technology implementation must be evaluated before implementation to verify the impact on data integrity.

5. Quality risk management

Note: documentation of data flows and data process maps are recommended to facilitate the assessment, mitigation and control of data integrity risks across the actual and intended data process(es).

5.1. Data Integrity Risk Assessment (DIRA) should be carried out in order to identify and assess areas of risk. This should cover systems and processes that produce data or, where data are obtained and inherent risks. The DIRAs should be risk-based, cover the life cycle of data and consider data criticality. Data criticality may be determined by considering how the data is used to influence the decisions made. The DIRAs should be documented and reviewed, as required, to ensure that it remains current.

5.2. The risk assessments should evaluate, for example, the relevant GxP computerised systems, supporting personnel, training, quality systems and outsourced activities.

5.3. DI risks should be assessed and mitigated. Controls and residual risks should be communicated. Risk review should be done throughout the document and data life cycle at a frequency based on the risk level, as determined by the risk assessment process.

5.4. Where the risk assessment has highlighted areas for remedial action, the prioritisation of actions (including the acceptance of an appropriate level of residual risk) and the prioritisation of controls should be documented and communicated. Where long-term remedial actions are identified, risk-reducing short-term measures should be implemented in order to provide acceptable data governance in the interim.

5.5. Controls identified may include organizational, procedural and technical controls such as procedures, processes, equipment, instruments and other systems in order to both prevent and detect situations that may impact on data integrity. Examples include the appropriate content and design of procedures, formats for recording, access control, the use of computerized systems and other means.

5.6. Efficient risk-based controls should be identified and implemented to address risks impacting data integrity. Risks include, for example, the deletion of, changes to and exclusion of data or results from data sets without written justification, authorisation where appropriate, and detection. The effectiveness of the controls should be verified (see Appendix 1 for examples).

6. Management review

6.1. Management should ensure that systems (such as computerized systems and paper systems) are meeting regulatory requirements in order to support data integrity compliance.

6.2. The acquisition of non-compliant computerized systems and software should be avoided. Where existing systems do not meet current requirements, appropriate controls should be identified and implemented based on risk assessment.

6.3. The effectiveness of the controls implemented should be evaluated through, for example:

- the tracking and trending of data;
- a review of data, metadata and audit trails (e.g. in warehouse and material management, production, quality control, case report forms and data processing); and
- routine audits and/or self-inspections, including data integrity and computerized systems.

7. Outsourcing

7.1. The selection of a contract acceptor should be done in accordance with an authorized procedure. The outsourcing of activities, ownership of data, and responsibilities of each party (contract giver and contract accepter) should be clearly described in written agreements. Specific attention should be given to ensuring compliance with data integrity requirements. Provisions should be made for responsibilities relating to data when an agreement expires.

7.2. Compliance with the principles and responsibilities should be verified during periodic site audits. This should include the review of procedures and data (including raw data and metadata, paper records, electronic data, audit trails and other related data) held by the relevant contract accepter identified in risk assessment.

7.3. Where data and document retention are contracted to a third party, particular attention should be given to security, transfer, storage, access and restoration of data held under that agreement, as well as controls to ensure the integrity of data over their life cycle. This includes static data and dynamic data. Mechanisms, procedures and tools should be identified to ensure data integrity and data confidentiality, for example, version control, access control, and encryption.

7.4. GxP activities, including outsourcing of data management, should not be sub-contracted to a third party without the prior approval of the contract giver. This should be stated in the contractual agreements.

7.5. All contracted parties should be aware of the requirements relating to data governance, data integrity and data management.

8. Training

8.1. All personnel who interact with GxP data and who perform GxP activities should be trained in relevant data integrity principles and abide by organization policies and procedures. This should include understanding the potential consequences in cases of non-compliance.

8.2. Personnel should be trained in good documentation practices and measures to prevent and detect data integrity issues.

8.3. Specific training should be given in cases where computerized systems are used in the generation, processing, interpretation and reporting of data and where risk assessment has shown that this is required to relevant personnel. Such training should include validation of computerized systems and for example, system security assessment, back-up, restoration, disaster recovery, change and configuration management, and reviewing of electronic data and metadata, such as audit trails and logs, for each GxP computerized systems used in the generation, processing and reporting of data.

9. Data, data transfer and data processing

9.1. Data may be recorded on paper or captured electronically by using equipment and instruments including those linked to computerised systems. A combination of paper and electronic formats may also be used, referred to as a "hybrid system".

9.2. Data integrity consideration are also applicable to media such as photographs, videos, DVDs, imagery and thin layer chromatography plates. There should be a documented rationale for the selection of such a method.

9.3. Risk-reducing measures such as scribes, second person oversight, verification and checks should be implemented where there is difficulty in accurately and contemporaneously recording data related to critical process parameters or critical quality attributes.

9.4. Results and data sets require independent verification if deemed necessary from the DIRA or by another requirement.

9.5. Programmes and methods (such as processing methods in sample analysis (see also Good Chromatography Practices, TRS 1025) should ensure that data meet ALCOA+ principles. Where results or data are processed using a different method/parameters, then each version of the processing method should be recorded. Data records, content versions together with audit trails containing the required details should allow for reconstruction of all data processing in GxP computerized systems over the data life cycle.

9.6. Data transfer/migration procedures should include a rationale and be robustly designed and validated to ensure that data integrity is maintained during the data life cycle. Careful consideration should be given to understanding the data format and the potential for alteration at each stage of data generation, transfer and subsequent storage. The challenges of migrating data are often underestimated, particularly regarding maintaining the full meaning of the migrated records.

Data transfer should be validated. The data should not be altered during or after it is transferred to the worksheet or other application. There should be an audit trail for this process. The appropriate quality procedures should be followed if the data transfer during the operation has not occurred correctly. Any changes in the middle layer software should be managed through the appropriate Quality Management Systems (*7*).

10. Good documentation practices

Note: The principles contained in this section are applicable to paper data.

10.1. Good documentation practices should be implemented and enforced to ensure compliance with ALCOA+ principles.

10.2. Data and recorded media should be durable. Ink should be indelible. Temperature-sensitive or photosensitive inks and other erasable inks should not be used. Where related risks are identified, means should be identified in order to ensure traceability of the data over their life cycle.

10.3. Paper should not be temperature-sensitive, photosensitive or easily oxidizable. If this is not feasible or limited, then true or certified copies should be generated.

10.4. Specific controls should be implemented in order to ensure the integrity of raw data and results recorded on paper records. These may include, but are not limited to:

- control over the issuance and use of loose paper sheets at the time of recording data;
- no use of pencil or erasers;
- use of single-line cross-outs to record changes with the identifiable person who made the change, date and reason for the change recorded (i.e. the paper equivalent to an electronic audit trail);
- no use of correction fluid or otherwise, obscuring the original record;
- controlled issuance of bound, paginated notebooks;
- controlled issuance and reconciliation of sequentially numbered copies of blank forms with authenticity controls;
- maintaining a signature and initial record for traceability and defining the levels of signature of a record; and
- archival of records by designated personnel in secure and controlled archives.

11. Computerized systems

(Note. This section highlights some specific aspects relating to the use of computerized systems. It is not intended to repeat the information presented in the other WHO guidelines here, such as the WHO Guideline on computerized systems (3), WHO Guideline on validation (2) and WHO Guideline on good chromatography practices (7). See references.)

11.1. Each computerized system selected should be suitable, validated for its intended use, and maintained in a validated state.

11.2. Where GxP systems are used to acquire, record, transfer, store or process data, management should have appropriate knowledge of the risks that the system and users may pose to the integrity of the data.

11.3. Software of computerized systems, used with GxP instruments and equipment, should be appropriately configured (where required) and validated. The validation should address for example the design, implementation and maintenance of controls in order to ensure the integrity of manually and automatically acquired data; ensure that Good Documentation Practices will be implemented; and that data integrity risks will be appropriately managed throughout the data life cycle. The potential for unauthorized and adverse manipulation of data during the life cycle of the data should be mitigated and, where possible, eliminated.

11.4. Where electronic instruments (e.g. certain pH meters, balances and thermometers) or systems with no configurable software and no electronic data retention are used, controls should be put in place to prevent the adverse manipulation of data and to prevent repeat testing to achieve the desired result.

11.5. Appropriate controls for the detection of lapses in data integrity principles should be in place. Technical controls should be used whenever possible but additional procedural or administrative controls should be implemented to manage aspects of computerised system control where technical controls are missing. For example, when stand-alone computerized systems with a user-configurable output are used, Fourier-transform infrared spectroscopy (FTIR) and UV spectrophotometers have user-configurable output or reports that cannot be controlled using technical controls. Other examples of non-technical detection and prevention mechanisms may include, but are not limited to, instrument usage logbooks and electronic audit trails.

Access and privileges

11.6. There should be a documented system in place that defines the access and privileges of users of systems. There should be no discrepancy between paper records and electronic records where paper systems are used to request changes for the creation and inactivation of users. Inactivated users should be retained in the system. A list of active and inactivated users should be maintained throughout the system life cycle.

11.7. Access and privileges should be in accordance with the role and responsibility of the individual with the appropriate controls to ensure data integrity (e.g. no modification, deletion or creation of data outside the defined privilege and in accordance with the authorized procedures defining review and approval where appropriate).

11.8. A limited number of personnel, with no conflict of interest in data, should be appointed as system administrators. Certain privileges such as data deletion, database amendment or system configuration changes should not be assigned to administrators without justification – and such activities should only be done with documented evidence of authorization by another responsible person. Records should be maintained and audit trails should be enabled in order to track activities of system administrators. As a minimum, activity logging for such accounts and the review of logs by designated roles should be conducted in order to ensure appropriate oversight.

11.9. For systems generating, amending or storing GxP data, shared logins or generic user access should not be used. The computerised system design should support individual user access. Where a computerised system supports only a single user login or limited numbers of user logins and no suitable alternative computerised system is available, equivalent control should be provided by third-party software or a paper-based method that provides traceability (with version control). The suitability of alternative systems should be justified and documented (*8*). The use of legacy hybrid systems should be discouraged and a priority timeline for replacement should be established.

Audit trail

11.10. GxP systems should provide for the retention of audit trails. Audit trails should reflect, for example, users, dates, times, original data and results, changes and reasons for changes (when required to be recorded), and enabling and disenabling of audit trails.

11.11. All GxP relevant audit trails should be enabled when software is installed and remain enabled at all times. There should be evidence of enabling the audit trail. There should be periodic verification to ensure that the audit trail remains enabled throughout the data life cycle.

11.12. Where a system cannot support ALCOA+ principles by design (e.g. legacy systems with no audit trail), mitigation measures should be taken for defined temporary periods. For example, add-on software or paper-based controls may be used. The suitability of alternative systems should be justified and documented. This should be addressed within defined timelines.

Electronic signatures

11.13. Each electronic signature should be appropriately controlled by, for example, senior management. An electronic signature should be:

- attributable to an individual;
- free from alteration and manipulation
- be permanently linked to their respective record; and
- date- and time-stamped.

11.14. An inserted image of a signature or a footnote indicating that the document has been electronically signed is not adequate unless it was created as part of the validated electronic signature process. The metadata associated with the signature should be retained.

Data backup, retention and restoration

11.15. Data should be retained (archived) in accordance with written policies and procedures, and in such a manner that they are protected, enduring, readily retrievable and remain readable throughout the records retention period. True copies of original records may be retained in place of the original record, where justified. Electronic data should be backed up according to written procedures.

11.16. Data and records, including backup data, should be kept under conditions which provide appropriate protection from deterioration. Access to such storage areas should be controlled and should be accessible only by authorized personnel.

11.17. Data retention periods should be defined in authorized procedures.

11.18. The decision for and manner in which data and records are destroyed, should be described in written procedures. Records for the destruction should be maintained.

11.19. Backup and restoration processes should be validated. The backup should be done routinely and periodically be restored and verified for completeness and accuracy of data and metadata. Where any discrepancies are identified, they should be investigated and appropriate action taken.

12. Data review and approval

12.2. There should be a documented procedure for the routine and periodic review, as well as the approval of data. Personnel with appropriate knowledge and experience should be responsible for reviewing and checking data. They should have access to original electronic data and metadata.

12.3. The routine review of GxP data and meta data should include audit trails. Factors such as criticality of the system (high impact versus low impact) and category of audit trail information (e.g. batch specific, administrative, system activities, and so on) should be considered when determining the frequency of the audit trail review.

12.4. A procedure should describe the actions to be taken where errors, discrepancies or omissions are identified in order to ensure that the appropriate corrective and preventive actions are taken.

12.5. Evidence of the review should be maintained.

12.6. A conclusion, where required, following the review of original data, metadata and audit trail records should be documented, signed and dated.

13. Corrective and preventive actions

13.1. Where organizations use computerized systems (e.g. for GxP data acquisition, processing, interpretation, reporting) which do not meet current GxP requirements, an action plan towards upgrading such systems should be documented and implemented in order to ensure compliance with current GxP.

13.2. When lapses in GxP relevant data regarding data integrity are identified, a risk-based approach may be used to determine the scope of the investigation, root cause, impact and CAPA, as appropriate. Health authorities, contract givers and other relevant organizations should be notified if the investigation identifies a significant impact or risk to, for example, materials, products, patients, reported information or data in application dossiers, and clinical trials.

References

1. Guidelines on good manufacturing practices for pharmaceutical products: main principle. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: forty-eighth report. Geneva: World Health Organization; 2013: Annex 2 (WHO Technical Report Series, No. 986; https://www.who.int/medicines/areas/quality_safety/quality_assurance/TRS986annex2.pdf?ua=1, accessed 4 May 2020).
2. Good manufacturing practices: guidelines on validation. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations; fifty-third report. Geneva: World Health Organization; 2019: Annex 3 (WHO Technical Report Series, No. 1019; http://digicollection.org/whoqapharm/documents/s23430en/s23430en.pdf, accessed 5 May 2020).
3. Good manufacturing practices: guidelines on validation. Appendix 5. Validation of computerized systems. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-third report. Geneva: World Health Organization; 2019: Annex 3 (WHO Technical Report Series, No. 1019; https://www.who.int/medicines/areas/quality_safety/quality_assurance/WHO_TRS_1019_Annex3.pdf?ua=1, accessed 4 May 2020).
4. Guidelines on quality risk management. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: forty-seventh report. Geneva: World Health Organization; 2013: Annex 2 (WHO Technical Report Series, No. 981; https://www.who.int/medicines/areas/quality_safety/quality_assurance/Annex2TRS-981.pdf, accessed 4 May 2020).
5. ICH harmonised tripartite guideline. Quality risk management Q9. Geneva: International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceutical for Human Use; 2005 (https://database.ich.org/sites/default/files/Q9%20Guideline.pdf, accessed 12 June 2020).
6. Good chromatography practices. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-fourth report. Geneva: World Health Organization; 2020: Annex 4 (WHO Technical Report Series, No. 1025; https://www.who.int/publications/i/item/978-92-4-000182-4, accessed 12 June 2020).
7. MHRA GxP data integrity guidance and definitions; Revision 1: Medicines & Healthcare Products Regulatory Agency (MHRA), London, March 2018 (https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/687246/MHRA_GxP_data_integrity_guide_March_edited_Final.pdf, accessed 12 June 2020).

Further reading

- Data integrity and compliance with CGMP guidance for industry: questions and answers guidance for industry. U.S. Department of Health and Human Services, Food and Drug Administration; 2016 (https://www.fda.gov/files/drugs/published/Data-Integrity-and-Compliance-With-Current-Good-Manufacturing-Practice-Guidance-for-Industry.pdf, accessed 15 June 2020).

- Good Practices for data management and integrity in regulated GMP/GDP environments. Pharmaceutical Inspection Convention and Pharmaceutical Inspection Co-operation Scheme (PIC/S), November 2018 (https://picscheme.org/layout/document.php?id=1567, accessed 15 June 2020).
- Baseline guide Vol 7: risk-based manufacture of pharma products; 2nd edition.
- ISPE Baseline® Guide, July 2017. ISPEGAMP® guide: records and data integrity; March 2017.
- Data integrity management system for pharmaceutical laboratories PDA Technical Report, No. 80; August 2018.
- ICH harmonised tripartite guideline. Pharmaceutical Quality System Q10. Geneva: International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceutical for Human Use; 2008 (https://database.ich.org/sites/default/files/Q10%20Guideline.pdf, accessed 2 October 2020).

Appendix 1

Examples in data integrity management

This Appendix reflects on some examples in data integrity management in order to support the main text on data integrity. It should be noted that these are examples and are intended for the purpose of clarification only.

Example 1: Quality risk management and data integrity risk assessment

Risk management is an important part of good practices (GxP). Risks should be identified and assessed and controls identified and implemented in order to assist manufacturers in preventing possible DI lapses.

As an example, a Failure Mode and Effects Analysis (FMEA) model (or any other tool) can be used to identify and assess the risks relating to any system where data are, for example, acquired, processed, recorded, saved and archived. The risk assessment can be done as a prospective exercise or retrospective exercise. Corrective and preventive action (CAPA) should be identified, implemented and assessed for its effectiveness.

For example, if during the weighing of a sample, the entry of the date was not contemporaneously recorded on the worksheet but the date is available on the print-out from a weighing balance and log book for the balance for that particular activity. The fact that the date was not recorded on the worksheet may be considered a lapse in data integrity expectations. When assessing the risk relating to the lack of the date in the data, the risk may be considered different (lower) in this case as opposed to a situation when there is no other means of traceability for the activity (e.g. no print-out from the balance). When assessing the risk relating to the lapse in data integrity, the severity could be classified as "low" (the data is available on the print-out); it does not happen on a regular basis (occurrence is "low"), and it could easily be detected by the reviewer (detection is "high") – therefore the overall risk factor may be considered low. The root cause as to why the record was not made in the analytical report at the time of weighing should still be identified and the appropriate action taken to prevent this from happening again.

Example 2: Good documentation practices in data integrity

Documentation should be managed with care. These should be appropriately designed in order to assist in eliminating erroneous entries, manipulation and human error.

Formats

Design formats to enable personnel to record or enter the correct information contemporaneously. Provision should be made for entries such as, but not limited to, dates, times (start and finish time, where appropriate), signatures, initials, results, batch numbers and equipment identification numbers. When a computerized system is used, the system should prompt the personnel to make the entries at the appropriate step.

Blank sheets of paper

The use of blank sheets should not be encouraged. Where blank sheets are used (e.g. to supplement worksheets, laboratory notebooks and master production and control records), the appropriate controls have to be in place and may include, for example, a numbered set of blank sheets issued which are reconciled upon completion. Similarly, bound paginated notebooks, stamped or formally issued by designated personnel, allow for the detection of unofficial notebooks and any gaps in notebook pages. Authorization may include two or three signatures with dates, for example, “prepared by” or “entered by”, “reviewed by” and “approved by”.

Error in recording data

Care should be taken when entries of data and results (electronic and paper records) are made. Entries should be made in compliance with good documentation practices. Where incorrect information had been recorded, this may be corrected provided that the reason for the error is documented, the original entry remains readable and the correction is signed and dated.

Example 3: Data entry

Data entry includes for example sample receiving registration, sample analysis result recording, logbook entries, registers, batch manufacturing record entries and information in case report forms. The recording of source data on paper records should be done using indelible ink, in a way that is complete, accurate, traceable, attributable and free from errors. Direct entry into electronic records should be done by responsible and appropriately trained individuals. Entries should be traceable to an individual (in electronic records, thus having an individual user access) and traceable to the date (and time, where relevant). Where appropriate, the entry should be verified by a second person or entered through technical means such as the scanning of bar-codes, where possible, for the intended use of these data. Additional controls may include the locking of critical data entries after the data are verified and a review of audit trails for critical data to detect if they have been altered. The manual entry of data from a paper record into a computerized system should be traceable to the paper records used which are kept as original data.

Example 4: Dataset

All data should be included in the dataset unless there is a documented, justifiable, scientific explanation and procedure for the exclusion of any result or data. Whenever out of specification or out of trend or atypical results are obtained, they should be investigated in accordance with written procedures. This includes investigating and determining CAPA for invalid runs, failures, repeats and other atypical data. The review of original electronic data should include checks of all locations where data may have been stored, including locations where voided, deleted, invalid or rejected data may have been stored. Data and metadata related to a particular test or product should be recorded together. The data should be appropriately stored in designated folders. The data should not be stored in other electronic folders or in other operating system logs. Electronic data should be archived in accordance with a standard operating procedure. It is important to ensure that associated metadata are archived with the relevant data set or securely traceable to the data set through relevant documentation. It should be possible to successfully retrieve all required data and metadata from the archives. The retrieval and verification should be done at defined intervals and in accordance with an authorized procedure.

Example 5: Legible and enduring

Data and metadata should be readable during the life cycle of the data. Electronic data are normally only legible/readable through the original software application that created it. In addition, there may be restrictions around the version of a software application that can read the data. When storing data electronically, ensure that any restrictions which may apply and the ability to read the electronic data are understood. Clarification from software vendors should be sought before performing any upgrade, or when switching to an alternative application, to ensure that data previously created will be readable.

Other risks include the fading of microfilm records, the decreasing readability of the coatings of optical media such as compact disks (CDs) and digital versatile/video disks (DVDs), and the fact that these media may become brittle.

Similarly, historical data stored on magnetic media will also become unreadable over time as a result of deterioration. Data and records should be stored in an appropriate manner, under the appropriate conditions.

Example 6: Attributable

Data should be attributable, thus being traceable to an individual and where relevant, the measurement system. In paper records, this could be done through the use of initials, full handwritten signature or a controlled personal seal. In electronic records, this could be done through the use of unique user logons that link the user to actions that create, modify or delete data; or unique electronic signatures which can be either biometric or non-biometric. An audit trail should capture user identification (ID), date and time stamps and the electronic signature should be securely and permanently linked to the signed record.

Example 7: Contemporaneous

Personnel should record data and information at the time these are generated and acquired. For example, when a sample is weighed or prepared, the weight of the sample (date, time, name of the person, balance identification number) should be recorded at that time and not before or at a later stage. In the case of electronic data, these should be automatically date- and time-stamped. In case hybrid systems are to be used, including the use for an interim period, the potential and criticality of system breaches should be covered in the assessment with documented mitigating controls in place. (The replacement of hybrid systems should be a priority with a documented CAPA plan.) The use of a scribe to record an activity on behalf of another operator should be considered only on an exceptional basis and should only take place where, for example, the act of recording places the product or activity at risk, such as, documenting line interventions by aseptic area operators. It needs to be clearly documented when a scribe has been applied.

> *"In these situations, the recording by the second person should be contemporaneous with the task being performed, and the records should identify both the person performing the task and the person completing the record. The person performing the task should countersign the record wherever possible, although it is accepted that this countersigning step will be retrospective. The process for supervisory (scribe) documentation completion should be described in an approved procedure that specifies the activities to which the process applies."* (Extract taken from the Medicines & Healthcare Products Regulatory Agency (MHRA) *GxP data integrity guidance and definitions* (*10*).)

A record of employees indicating, their name, signature, initials or other mark or seal used should be maintained to enable traceability and to uniquely identify them and the respective action.

Example 8: Changes

When changes are made to any GxP result or data, the change should be traceable to the person who made the change as well as the date, time and reason for the change. The original value should not be obscured. In electronic systems, this traceability should be documented via computer generated audit trails or in other metadata fields or system features that meet these requirements. Where an existing computerized system lacks computer-generated audit trails, personnel may use alternative means such as procedurally controlled use of logbooks, change control, record version control or other combinations of paper and electronic records to meet GxP regulatory expectations for traceability to document the what, who, when and why of an action.

Example 9: Original

The first or source capture of data or information and all subsequent data required to fully reconstruct the conduct of the GxP activity should be available. In some cases, the electronic data (electronic chromatogram acquired through high-performance liquid chromatography (HPLC)) may be the first source of data and, in other cases, the recording of the temperature on a log sheet in a room – by reading the value on a data logger. This data should be reviewed according to the criticality and risk assessment.

Example 10: Controls

Based on the outcome of risk assessment which should cover all areas of data governance and data management, appropriate and effective controls should be identified and implemented in order to assure that all data, whether in paper records or electronic records, will meet GxP requirements and ALCOA+ principles. Examples of controls may include, but are not limited to:

- the qualification, calibration and maintenance of equipment, such as balances and pH meters, that generate printouts;
- the validation of computerized systems that acquire, process, generate, maintain, distribute, store or archive electronic records;
- review and auditing of activities to ensure that these comply with applicable GxP data integrity requirements;
- the validation of systems and their interfaces to ensure that the integrity of data will remain while transferring between/among computerized systems;
- evaluation to ensure that computerized systems remain in a validated state;
- the validation of analytical procedures;

- the validation of production processes;
- a review of GxP records;
- ensuring effective review and oversight of the Batch Release Systems and processes by using different oversight and review techniques to ensure that data have not changed since the original entry; and
- the investigation of deviations, out of trend and out of specifications results.

Example 11: Accuracy

Points to consider for assuring accurate GxP records:

- the entry of critical data into a computer by an authorized person (e.g. entry of a master processing formula) requires an additional check on the accuracy of the data entered manually. This check may be done by independent verification and release for use by a second authorized person or by validated electronic means. For example, to detect and manage risks associated with critical data, procedures would require verification by a second person;
- validation and control over formulae for calculations including electronic data capture systems;
- ensuring correct entries into the laboratory information management system (LIMS) such as fields for specification ranges;
- other critical master data, as appropriate. Once verified, these critical data fields should normally be locked in order to prevent further modification and only be modified through a formal change control process;
- the process of data transfer between systems should be validated;
- the migration of data including planned testing, control and validation; and
- when the activity is time-critical, printed records should display the date and time stamp.